

Aakash Panchagnula

Email: aakash.panchagnula@gmail.com | Phone: 412-844-1154 | [LinkedIn](#) | [GitHub](#)

Machine Learning & DevOps Engineer with 2+ years of experience building scalable backend systems and automating ML pipelines using Python, Node.js, and AWS. Proficient in deploying and managing production-grade machine learning models with SageMaker, EMR, and Spark, and integrating them through RESTful APIs. Skilled in cloud-native development, CI/CD automation, and container orchestration using Docker and Kubernetes. Experienced in designing distributed systems, optimizing data pipelines, and monitoring ML workflows with tools like Airflow, CloudWatch, and Git-based version control.

EDUCATION

Carnegie Mellon University

Master of Information Systems Management

Pittsburgh, PA

August 2023 - December 2024

Coursework: Object Oriented Programming, Machine Learning for Problem Solving, Distributed Systems

Data Focused Python, Unstructured Data Analytics, Advanced AI and Business Strategy, Statistics for IT Managers, Database Management

NMIMS University

Bachelor of Technology - Robotics Engineering

Mumbai, India

June 2017 - May 2021

Minor: Artificial Intelligence and Machine Learning

SKILLS

Languages: Python, JavaScript (Node.js), Java, C, C++, SQL, R, Bash

Frameworks & Libraries: Flask, FastAPI, Express.js, TensorFlow, PyTorch, scikit-learn, XGBoost, LightGBM, ML.NET

Machine Learning & MLOps: Model Deployment (AWS SageMaker, EMR), Feature Engineering, Hyperparameter Tuning, Ensemble Methods, A/B

Testing, Model Monitoring

Cloud & DevOps: AWS (S3, Lambda, ECS, SageMaker, EMR, Glue, Kinesis), Docker, Kubernetes, CI/CD (GitHub Actions, GitLab, Bitbucket Pipelines), CloudWatch

Big Data & Pipelines: Apache Spark, Hadoop, Apache Airflow, Kafka

Databases: PostgreSQL, MySQL, MongoDB, Cassandra, Snowflake, Redshift, BigQuery

Development Tools & IDEs: Git, GitHub, Bitbucket, VS Code, Postman, Jupyter Notebook

Statistical & Optimization Methods: Regression, Classification, Clustering, Time Series, PCA, Optimization (Linear, Integer), Anomaly Detection

Certifications: IBM Applied AI, IBM Advanced Machine Learning and Signal Processing, Google Cloud Sequence Models

WORK EXPERIENCE

Hyphenova Network

West Hollywood, CA

Software Engineer - MLOps

May 2024 - August 2024

- Automated deployment of ML pricing models using AWS SageMaker and Spark (on EMR), reducing model iteration time from **18 hours to 13 hours** (**≈28% decrease**)

- Developed Flask-based API endpoints for real-time model inference, integrated with AWS Kinesis and Lambda, cutting prediction latency from **950ms to 620ms** (**35% improvement**)

- Containerized versioned ML models with Docker and deployed via ECS with auto-scaling policies, increasing production uptime from **93% to 98%** over 3 months

- Created CI/CD pipelines for model retraining and deployment using GitHub Actions and integrated monitoring via CloudWatch alarms, reducing manual deployment work by **~40%**

Deloitte

Mumbai, India

Software Engineer - DevOps

May 2021 - July 2023

- Implemented automated infrastructure provisioning for 9 enterprise projects using AWS CloudFormation and Bitbucket Pipelines, reducing setup time by **45%** and onboarding time by **30%**

- Spearheaded SAP Ariba cloud migration for a \$15M retail client, boosting query response speed by **31%** and reducing server-side timeout errors by **22%** post-migration

- Re-architected backend ETL pipelines in SQL and Python for high-volume data (100M+ records/month), decreasing monthly batch processing time from **11 hours to 8.2 hours** (**~26% gain**)

- Set up centralized observability with AWS CloudWatch Logs, custom metrics, and alarms, reducing average incident resolution time from **3.5 hours to 2.3 hours** (**≈33% faster recovery**)

ACADEMIC PROJECTS

Geospatial Crime Mapping and Analytics using Big Data

May 2024 - August 2024

- Built a distributed processing pipeline with Hadoop MapReduce and Apache Spark to analyze **10 years of Pittsburgh crime data (~25K records)**, reducing runtime by **40%**

- Implemented spatial clustering using Haversine distance and DBSCAN, improving geographic crime pattern detection accuracy by **30%**

- Designed and served output in GeoJSON format for integration with Google Earth, enabling interactive visual mapping for law enforcement use

- Tuned Spark job configurations for memory optimization, resulting in a **22% performance improvement** during high-volume batch runs

Anomaly Detection for AirBnb Listings

August 2023 - December 2023

- Built a custom anomaly detection module for Airbnb property data (Los Angeles, ~80K listings) using Isolation Forest and One-Class SVM

- Detected and filtered inconsistent property metrics with **92% precision**, improving downstream model quality and pricing recommendation accuracy

- Integrated Singular Spectrum Analysis (SSA) spike detection using ML.NET for time-based outlier detection, boosting temporal anomaly capture by **~18%**

- Deployed the model as a callable function within ETL jobs, improving QA pipeline efficiency by **~25%**